Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [14ME71]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2018

MECHANICAL VIBRATIONS

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer any **FIVE** full Questions choosing atleast two from each part.
ii. Missing data may be suitably assumed.

PART - A

1. a. With a sketch, explain beats phenomenon and obtain its resultant motion. **04**

   b. Split the harmonic motion $x = 10\sin\left(wt + \frac{\pi}{6}\right)$, into two harmonic motions one having a phase angle of zero and other 45°. **04**

   c. Periodic motion in time domain is given by $x(t) = -20t + 2 \;\; for\; 0 \le 2$. Obtain Fourier's series equation in frequency domain. **06**

2. a. Determine the effect of mass of the spring on the natural frequency of the spring mass system. **04**

   b. A spring mass system has a stiffness of 'K' N/m and a mass 'M' Kg. It has natural frequency of vibrations of 12cps. An extra 2Kg mass is coupled to mass 'M' and its net frequency becomes 10cps. Find 'K' and 'M'. **04**

   c. Use the Energy method to find the natural frequency of oscillation of the homogeneous cylinder shown in figure 2c. **06**

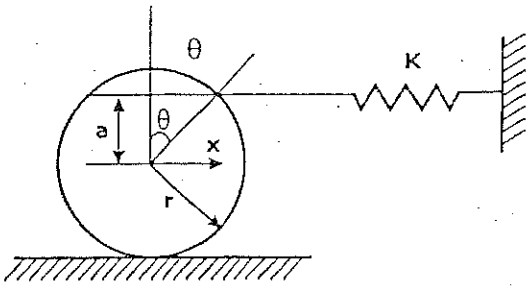

Fig.2c

3. a. Obtain the response of viscous damped system for under damped case. **04**

   b. The ratio of successive amplitudes of viscously damped single degree of freedom system is found to be 18:1. Determine the ratio of successive amplitudes if the amount of damping is halved. **04**

   c. Find the equation of motion for the system shown in figure 3c when (a) $\varepsilon = 1, b)\varepsilon = 0.3,\; c)\varepsilon = 2,$ if the mass m is displaced by a distance of 30mm and released. **06**

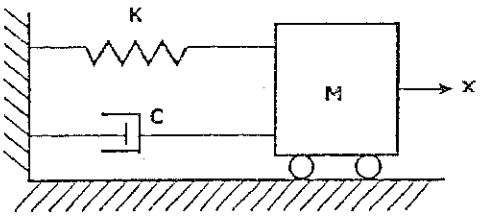

Fig. 3(c)

4. a. Define "Transmissibility". Derive an expression for "motion transmissibility" **05**

b. A vibrating body is supported by 6 isolators each having stiffness of 32000N/m and 6 dashpots each having 400 N-s/m. The vibrating body is to be isolated by a rotating device having an amplitude of 0.06mm at 600pm. Take m=30Kg. Determine the amplitude of vibration of the body and dynamic load on each isolator. **09**

PART-B

5. a. Write a brief note on Vibrometer. **04**

b. A commercial vibration pick-up has a natural frequency of 5.75CPS and a damping factor of 0.65. What is the lowest frequency beyond which the amplitude can be measured within $\pm 1\%$ error? **05**

c. A rotor having a mass of 5Kg is mounted midway on a 1Cm diameter shaft supported at ends by bearings. Bearings span is 40Cm. Eccentricity is 0.02mm. If the system rotates at 300 rpm find the amplitude of steady state vibrations and dynamic force transmitted to bearings. Take $E=1.96 \times 10^{11}$ N/m^2 and neglect damping. **05**

6. a. Determine the two natural frequencies and the corresponding mode shapes for the system shown in figure 6a. The string is stretched with a large tension T. Sketch the mode shapes. **09**

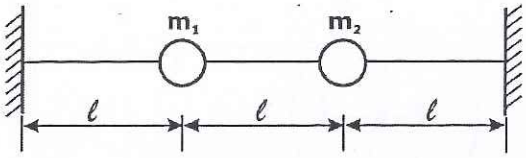

Fig. 6.(a)

b. A section of pipe pertaining to a certain machine vibrates with large amplitude at a compressor speed of 220rpm. For analyzing this system, a spring-mass system was suspended from the pipe to act as an absorber. A one Kg absorber mass tuned to 220cpm resulted in two resonant frequencies of 188 and 258cpm. What must be the mass and the spring stiffness of the absorber if the resonant frequencies are to be outside the range of 150 to 310cpm? **05**

7. a. Use the stodola method to determine the lowest natural frequency of a 4 – DOF spring mass system shown in figure 7a. **10**

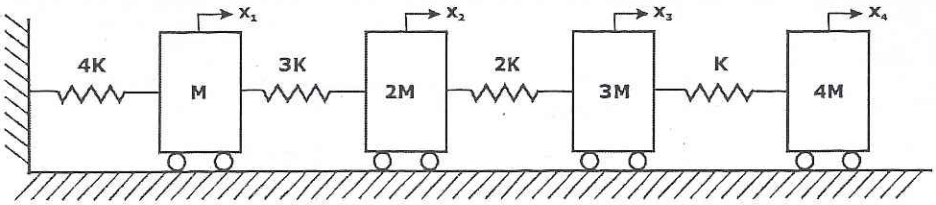

Fig. 7a

b. Determine the influence co-efficient of the system shown in figure-7b. 04

Take, $E = 2X10^{11} N/m^2$, $I = 4X10^{-7} m^4$.

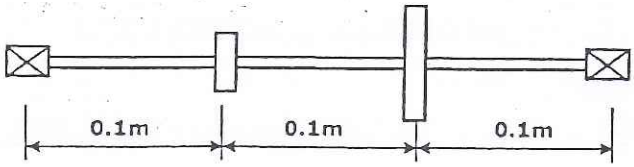

Fig. 7b

8. a. A rod carries a disc at its lower end as shown in figure 8a. Derive the expression for the natural frequencies. 07

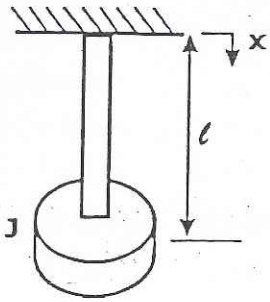

Fig. 8a

b. Derive an expression for the free longitudinal vibration of a uniform bar of length L, one end of which is fixed and the other end is free. 07